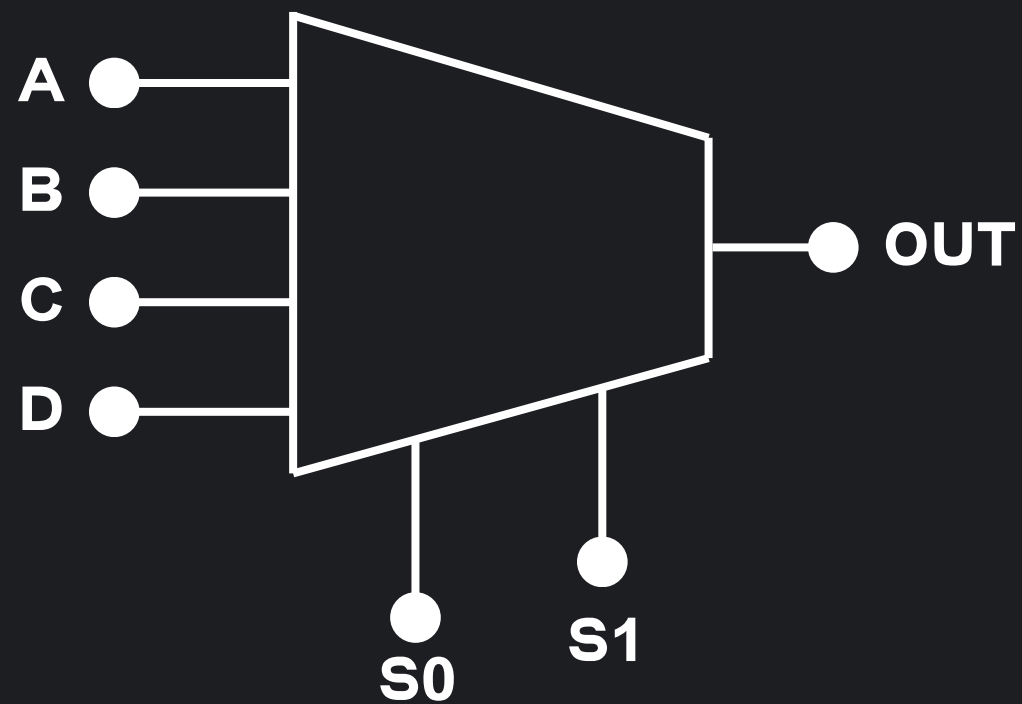


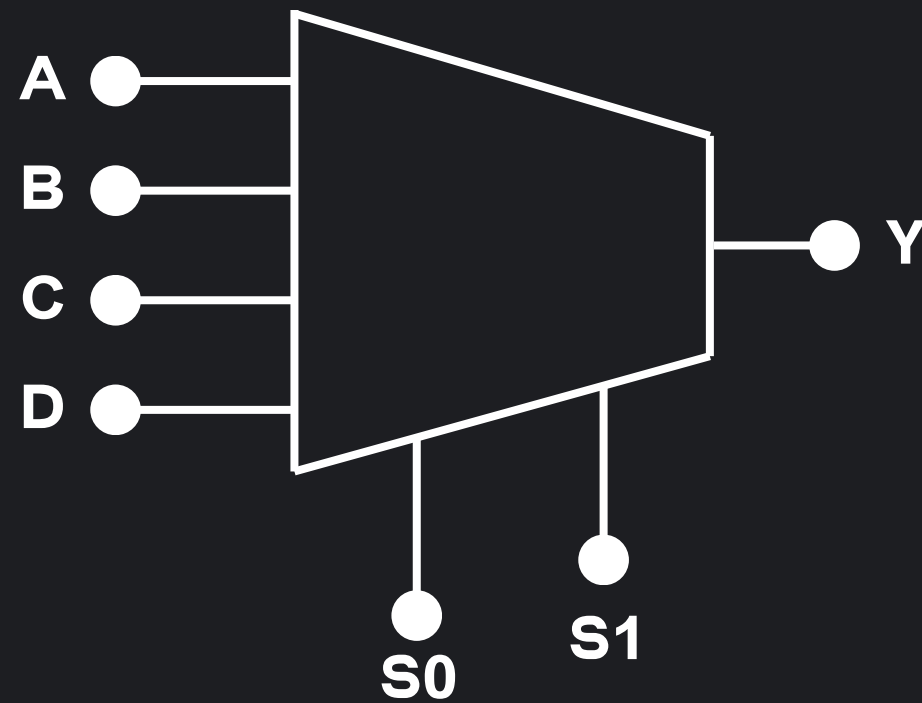
MULTIPLEXER 4 TO 1



What is a MULTIPLEXER?

A Multiplexer (MUX) is a basic combinational logic element that selects one signal from multiple inputs and passes it to a single output based on Selector inputs. The output is in a HIGH (1) state if the selected input is HIGH, while it is in a LOW (0) state if the selected input is LOW.

Truth Table



S1	S0	Y
0	0	A
0	1	B
1	0	C
1	1	D

Function of the MULTIPLEXER

A Multiplexer performs a logical data selection operation on its multiple inputs.

The main function of a Multiplexer is to serve as a digital selector switch that routes one of several inputs (A, B, C, D) to a single output (Y) according to this rule:

Output Y = A when Selectors S1_S0 = 00.

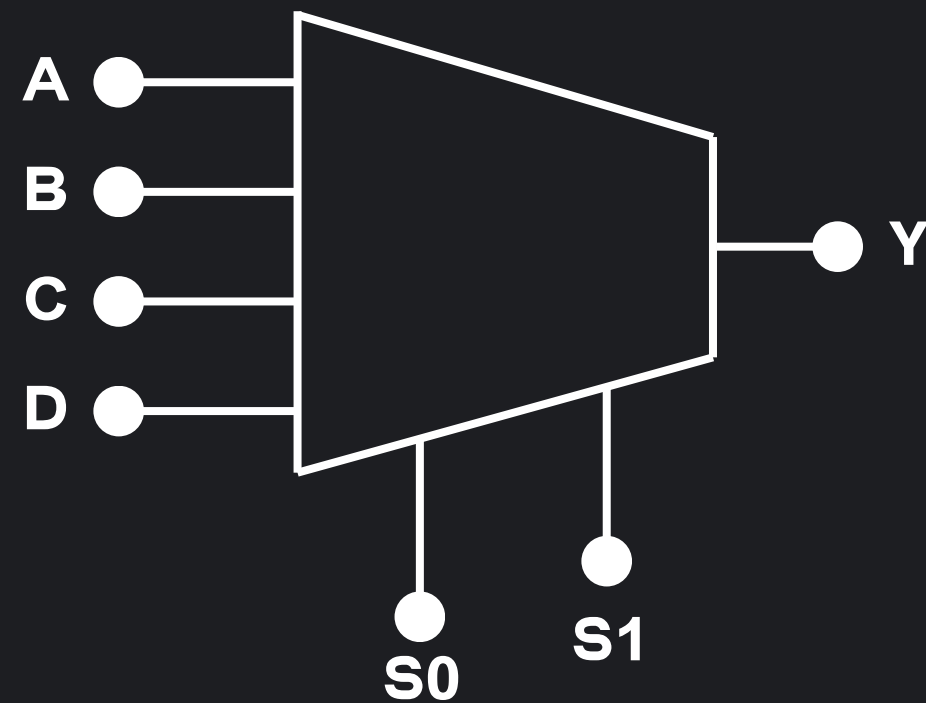
Output Y = B when Selectors S1_S0 = 01.

Output Y = C when Selectors S1_S0 = 10.

Output Y = D when Selectors S1_S0 = 11.

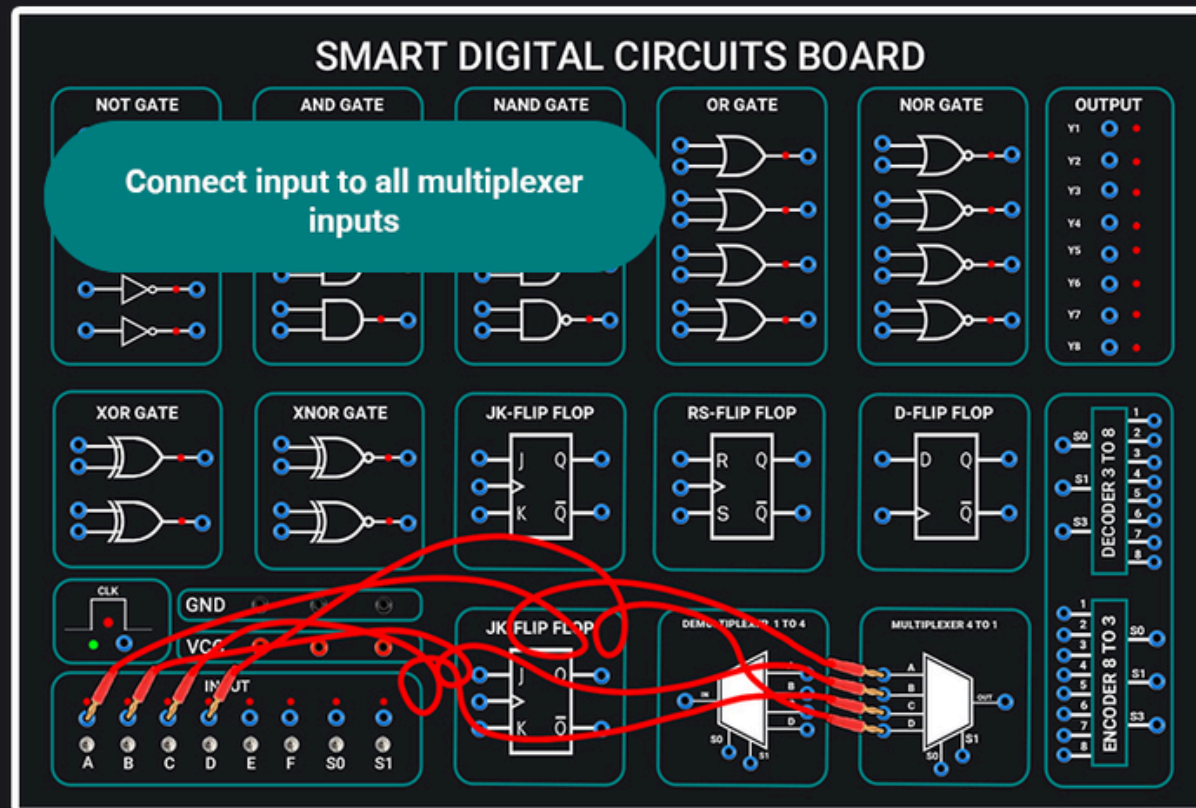
Q\ how to connect a MULTIPLEXER, write the Truth Table.

Sol:

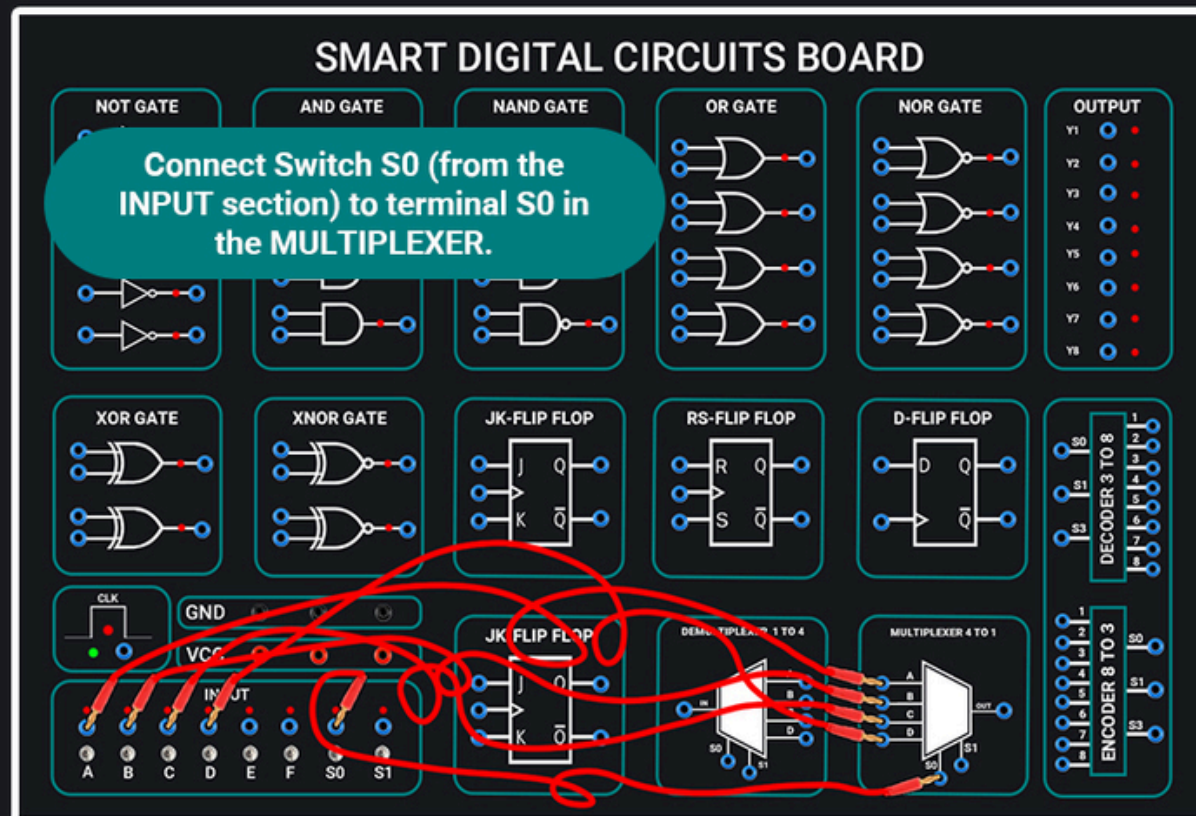


S1	S0	Y
0	0	A
0	1	B
1	0	C
1	1	D

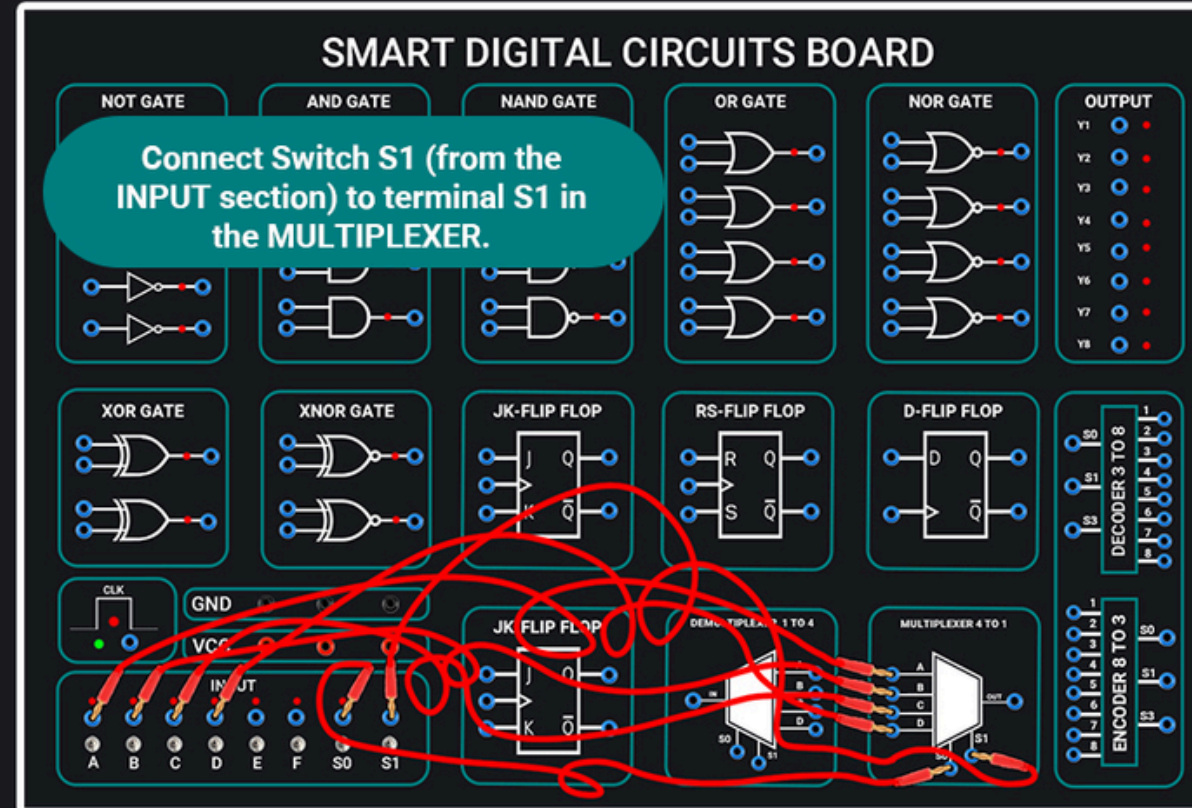
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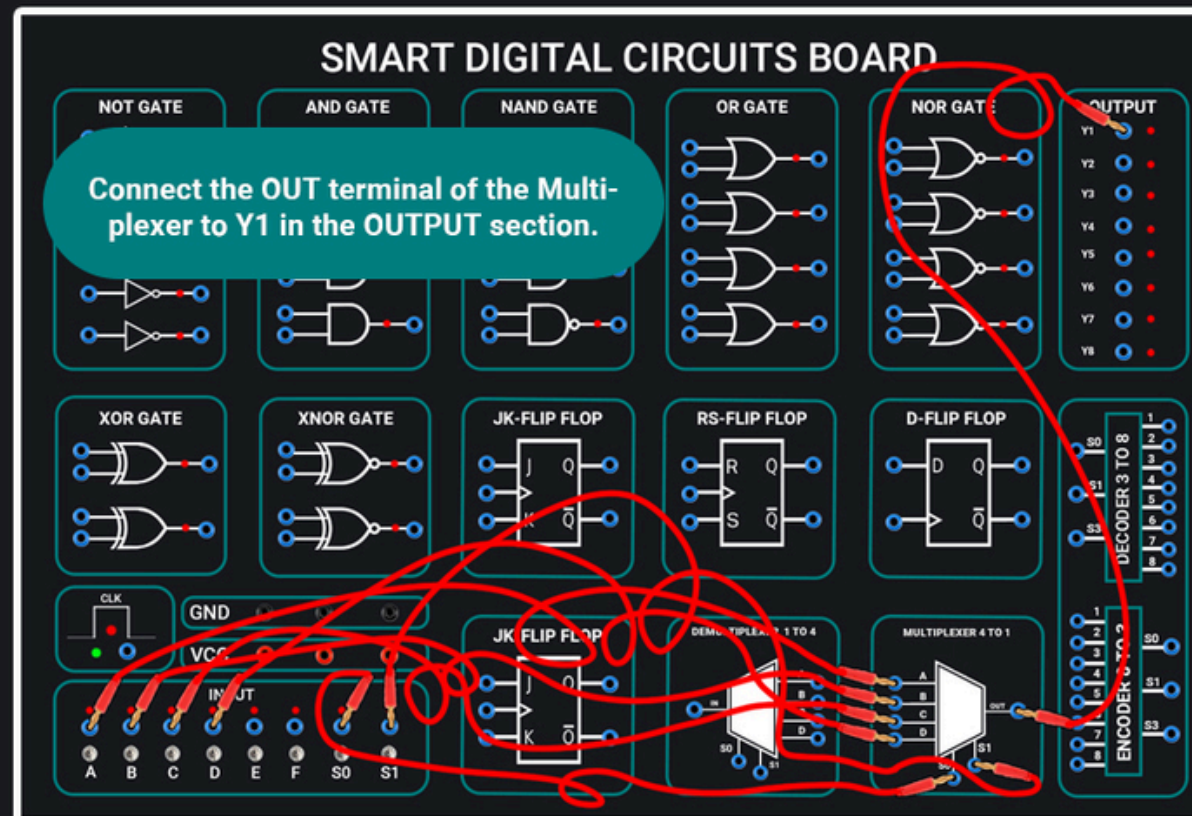
2



3



4



SMART DIGITAL CIRCUITS BOARD

